Skim Recap — backend comparison

LiteRT-LM (Gemma 4 E4B, 2.97 GB) against Chrome's built-in Prompt API, on the same prompts the extension ships.

Chrome	Chrome/151.0.0.0
Extension	0.4.0
LiteRT-LM · Gemma 4 E4B	available

litert · detail


```
{
 "model": "gemma-4-E4B-it-web.litertlm",
 "engineSettings": "engine not loaded yet — run once to populate",
 "cachedModelBytes": 2969059328,
 "gpuAdapter": {
  "vendor": "apple",
  "architecture": "metal-3",
  "device": "",
  "description": ""
 }
}
```


Chrome Prompt API	available
output languages	en · ja · es · de · fr · zh · NotSupportedError · zh-Hant · NotSupportedError · ko · NotSupportedError
input languages (page)	en · ja · es · de · fr · zh · NotSupportedError · zh-Hant · NotSupportedError · ko · NotSupportedError

prompt-api · detail


```
{
 "present": true,
 "streamSemantics": "unknown",
 "sessionSampling": {
  "temperature": 0,
  "topK": 1
 },
 "params": {
  "defaultTopK": 64,
  "maxTopK": 128,
  "defaultTemperature": 1,
  "maxTemperature": 2
 },
 "contextWindow": 9216,
 "inputQuota": 9216
}
```

Exploring Claude/GPT Knowledge Cutoffs & Pre-training Timelines RECAP → PAGE LANGUAGE

(unclassified capture)

LiteRT-LM · Gemma 4 E4B

RUN 1

Anthropic models Opus 4.7 onwards share a training run that cuts off around late December 2025.
The analysis uses an 8-way multiple choice quiz on daily Wikipedia facts to estimate knowledge cutoffs.
OpenAI models show a pattern of training on past model outputs, including GPT-4 and GPT-5.
Anthropic models show a pattern of training on past model outputs, including 3.5 Sonnet and Sonnet 4.5.

789 ms to first token · 3219 ms · 394 chars · 4 lines

RUN 2

Anthropic models Opus 4.7 onwards share a training run that cuts off around late December 2025.
The analysis uses an 8-way multiple choice quiz on daily Wikipedia facts to estimate knowledge cutoffs.
OpenAI models show a pattern of training on past model outputs, including GPT-4 and GPT-5.
Anthropic models show a pattern of training on past model outputs, including 3.5 Sonnet and Sonnet 4.5.

779 ms to first token · 3207 ms · 394 chars · 4 lines

identical across runs

Chrome Prompt API

RUN 1

Anthropic's Opus 4.7 models were trained on a dataset that cuts off around late December 2025.
Model knowledge cutoffs correlate with the dataset timespan used during pre-training.
Models often self-report identities based on frequent mentions of those identities in their training data.
Labs frequently train new models on outputs from previous versions of their models, potentially from user chats.

718 ms to first token · 2021 ms · 400 chars · 4 lines

RUN 2

Anthropic's Opus 4.7 models were trained on a dataset that cuts off around late December 2025.
Model knowledge cutoffs correlate with the dataset timespan used during pre-training.
Models often self-report identities based on frequent mentions of those identities in their training data.
Labs frequently train new models on outputs from previous versions of their models, potentially from user chats.

666 ms to first token · 1946 ms · 400 chars · 4 lines

identical across runs

Exploring Claude/GPT Knowledge Cutoffs & Pre-training Timelines FEYNMAN → PAGE LANGUAGE

(unclassified capture)

LiteRT-LM · Gemma 4 E4B

RUN 1

The passage explains how large language models, like Claude and GPT, are built through a multi-stage process. The most expensive part is "pre-training," which involves feeding the model massive amounts of data to build a foundational knowledge base. After this, teams perform "post-training" to refine the model's reasoning and problem-solving skills, often creating smaller, specialized versions of the main model.

To figure out when a model's knowledge stops—its "knowledge cutoff"—the author tested the models on recent facts from Wikipedia. By seeing when the models start getting confused or giving wrong answers about current events, the author could estimate the date the training data ended.

The author also looked at what the models claim to be when asked about their identity. If a model frequently says it is a specific version, it suggests that the developers trained it on a lot of data where that model's name was mentioned, likely from user conversations on the company's public chat websites.

816 ms to first token · 5753 ms · 1010 chars · 3 lines

RUN 2

The passage explains how large language models, like Claude and GPT, are built through a multi-stage process. The most expensive part is "pre-training," which involves feeding the model massive amounts of data to build a foundational knowledge base. After this, teams perform "post-training" to refine the model's reasoning and problem-solving skills, often creating smaller, specialized versions of the main model.

To figure out when a model's knowledge stops—

Chrome Prompt API

RUN 1

The passage discusses how large language models like Claude and GPT are developed, focusing on when they were initially trained and what that tells us about their knowledge. The initial, extensive training phase is called "pre-training," and it involves feeding the model massive amounts of data. A "checkpoint" is a snapshot of the model at a specific point during this pre-training process. The author estimates these checkpoints by testing the models on factual questions from Wikipedia to see when they start losing information from their training data.

After pre-training, models undergo "post-training," which involves further refinement to improve reasoning and problem-solving. This post-training can involve creating smaller versions of the model, a process called "distillation." The author suggests that the date of the pre-training dataset is closely linked to when a model's knowledge cutoff occurs – the point after which the model's information becomes outdated. The author uses a quiz method to estimate these cutoff dates.

Finally, the passage explores how models identify themselves. The more a model encounters phrases like "I am X" during training, the more likely it is to claim that identity. The author analyzes how models describe themselves in responses to prompts like "What model are you?" and finds patterns suggesting that labs often train on conversations from their own chatbots (like ChatGPT or Claude) to improve the models' ability to mimic those chatbots. This training on user interactions is likely embedded within the model's system prompt.

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The author also looked at what the models claim to be when asked about their identity. If a model frequently says it is a specific version, it suggests that the developers trained it on a lot of data where that model's name was mentioned, likely from user conversations on the company's public chat websites.

824 ms to first token · 5736 ms · 1010 chars · 3 lines

identical across runs

707 ms to first token · 5430 ms · 1588 chars · 3 lines

RUN 2

The passage discusses how large language models like Claude and GPT are developed, focusing on when they were initially trained and what that tells us about their knowledge. The initial, extensive training phase is called "pre-training," and it involves feeding the model massive amounts of data. A "checkpoint" is a snapshot of the model at a specific point during this pre-training process. The author estimates these checkpoints by testing the models on factual questions from Wikipedia to see when they start losing information from their training data.

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692 ms to first token · 5330 ms · 1588 chars · 3 lines

identical across runs

5.2.3. Saxagliptin RECAP → PAGE LANGUAGE (unclassified capture)

LiteRT-LM · Gemma 4 E4B

RUN 1

Saxagliptin inhibits the DPP-4 enzyme to prolong incretin hormone activity.
CYP3A4 variants can reduce saxagliptin clearance by 1.9–7%.
Canagliflozin inhibits SGLT2 to increase glucosuria and improve glycemic control.
Canagliflozin reduces systolic blood pressure by 3.93 mmHg and body weight by 1.6 kg.
Approximately 60% of canagliflozin is excreted in feces.

782 ms to first token · 3327 ms · 360 chars · 5 lines

RUN 2

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Approximately 60% of canagliflozin is excreted in feces.

802 ms to first token · 3347 ms · 360 chars · 5 lines

identical across runs

Chrome Prompt API

RUN 1

Saxagliptin inhibits DPP-4, prolonging incretin hormone activity to improve glucose homeostasis.
CYP3A4 metabolism of saxagliptin varies between variants, potentially affecting drug exposure and adverse effects.
Canagliflozin inhibits SGLT2, increasing glucosuria and improving glycemic control independently of insulin secretion.
Canagliflozin demonstrates cardiovascular and renal protective effects through mechanisms beyond glucose lowering, including blood pressure and weight reductions.

1049 ms to first token · 2500 ms · 493 chars · 4 lines

RUN 2

Saxagliptin inhibits DPP-4, prolonging incretin hormone activity to improve glucose homeostasis.
CYP3A4 metabolism of saxagliptin varies between variants, potentially affecting drug exposure and adverse effects.
Canagliflozin inhibits SGLT2, increasing glucosuria and improving glycemic control independently of insulin secretion.
Canagliflozin demonstrates cardiovascular and renal protective effects through mechanisms beyond glucose lowering, including blood pressure and weight reductions.

667 ms to first token · 2101 ms · 493 chars · 4 lines

identical across runs

5.2.3. Saxagliptin FEYNMAN → PAGE LANGUAGE (unclassified capture)

LiteRT-LM · Gemma 4 E4B

RUN 1

Saxagliptin works by blocking an enzyme called DPP-4, which normally breaks down helpful hormones like GLP-1 and GIP. These hormones help the body manage blood sugar by telling the pancreas to release insulin when needed and telling the liver to stop making too much sugar. By keeping these hormones active longer, saxagliptin helps lower blood sugar after meals and while fasting.

The way the body processes saxagliptin is heavily influenced by a specific liver enzyme called CYP3A4. Variations in this enzyme can cause the drug to be cleared from the body more slowly, potentially leading to higher drug levels and a greater risk of low blood sugar. Because of this, certain foods or supplements that slow down this enzyme can increase the amount of saxagliptin in the bloodstream.

Canagliflozin is a different type of diabetes drug that works by blocking the kidneys from reabsorbing sugar, causing more sugar to be excreted in the urine. Unlike some other drugs, it lowers blood sugar without significantly increasing the risk of hypoglycemia (dangerously low blood sugar). It also offers extra benefits, such as lowering blood pressure and improving kidney health.

799 ms to first token · 6525 ms · 1171 chars · 3 lines

RUN 2

Saxagliptin works by blocking an enzyme called DPP-4, which normally breaks down helpful hormones like GLP-1 and GIP. These hormones help the body manage blood sugar by telling the pancreas to release insulin when needed and telling the liver to stop making too much sugar. By keeping these hormones active longer, saxagliptin

Chrome Prompt API

RUN 1

Saxagliptin works by blocking an enzyme called DPP-4, which normally breaks down hormones called incretins. These incretins are important for controlling blood sugar by signaling the pancreas to release insulin when blood sugar is high and suppressing the release of glucagon, another hormone that raises blood sugar. By preventing DPP-4 from breaking down incretins, saxagliptin helps increase insulin release and lower glucose production in the liver, leading to better blood sugar control.

The way saxagliptin is processed in the body involves an enzyme called CYP3A4. Some people have variations in this enzyme that make it work slower, which can lead to higher levels of saxagliptin in the bloodstream. This could potentially increase the risk of side effects like hypoglycemia (low blood sugar), especially when taking saxagliptin with other medications. Certain foods and supplements can also affect CYP3A4, potentially altering how saxagliptin is metabolized.

Canagliflozin is another type of medication for type 2 diabetes that works differently. It inhibits SGLT2, a protein in the kidneys that normally reabsorbs glucose. By blocking SGLT2, canagliflozin causes the kidneys to excrete more glucose in the urine, which lowers blood sugar. It also has benefits beyond just lowering blood sugar, such as improving cardiovascular and kidney health. Canagliflozin is eliminated from the body mainly through the feces, with a smaller amount excreted in the urine.

720 ms to first token · 5513 ms · 1470 chars · 3 lines

RUN 2

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810 ms to first token · 6554 ms · 1171 chars · 3 lines

identical across runs

called incretins. These incretins are important for controlling blood sugar by signaling the pancreas to release insulin when blood sugar is high and suppressing the release of glucagon, another hormone that raises blood sugar. By preventing DPP-4 from breaking down incretins, saxagliptin helps increase insulin release and lower glucose production in the liver, leading to better blood sugar control.

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716 ms to first token · 5478 ms · 1470 chars · 3 lines

identical across runs

5.2.1. Repaglinide RECAP → PAGE LANGUAGE (unclassified capture)

LiteRT-LM · Gemma 4 E4B

RUN 1

Troglitazone, linagliptin, and pioglitazone are oral antidiabetic agents that are substrates of CYP3A4.
Repaglinide stimulates insulin secretion by inhibiting ATP-sensitive potassium channels on pancreatic β-cells.
Gemfibrozil inhibits both CYP2C8 and the hepatic uptake transporter OATP1B1, increasing repaglinide exposure eight-fold.
Nateglinide has a faster onset and shorter duration of action than repaglinide.

919 ms to first token · 3512 ms · 415 chars · 4 lines

RUN 2

Troglitazone, linagliptin, and pioglitazone are oral antidiabetic agents that are substrates of CYP3A4.
Repaglinide stimulates insulin secretion by inhibiting ATP-sensitive potassium channels on pancreatic β-cells.
Gemfibrozil inhibits both CYP2C8 and the hepatic uptake transporter OATP1B1, increasing repaglinide exposure eight-fold.
Nateglinide has a faster onset and shorter duration of action than repaglinide.

920 ms to first token · 3575 ms · 415 chars · 4 lines

identical across runs

Chrome Prompt API

RUN 1

Troglitazone, linagliptin, and pioglitazone are oral antidiabetic agents metabolized by CYP3A4, sometimes in conjunction with CYP2C8 or CYP2C9.
Repaglinide stimulates insulin secretion by inhibiting ATP-sensitive potassium channels, leading to calcium influx and insulin release.
CYP2C8 and CYP3A4 metabolize repaglinide, and interactions with drugs like clopidogrel and gemfibrozil can significantly alter repaglinide plasma concentrations, potentially increasing hypoglycemia risk.
Nateglinide, another meglitinide, promotes glucose-dependent insulin secretion via ATP-sensitive potassium channels, but has a faster onset and shorter duration of action than repaglinide.

1098 ms to first token · 3563 ms · 672 chars · 4 lines

RUN 2

Troglitazone, linagliptin, and pioglitazone are oral antidiabetic agents metabolized by CYP3A4, sometimes in conjunction with CYP2C8 or CYP2C9.
Repaglinide stimulates insulin secretion by inhibiting ATP-sensitive potassium channels, leading to calcium influx and insulin release.
CYP2C8 and CYP3A4 metabolize repaglinide, and interactions with drugs like clopidogrel and gemfibrozil can significantly alter repaglinide plasma concentrations, potentially increasing hypoglycemia risk.
Nateglinide, another meglitinide, promotes glucose-dependent insulin secretion via ATP-sensitive potassium channels, but has a faster onset and shorter duration of action than repaglinide.

721 ms to first token · 3246 ms · 672 chars · 4 lines

	`identical across runs`

5.2.1. Repaglinide FEYNMAN → PAGE LANGUAGE (unclassified capture)

LiteRT-LM · Gemma 4 E4B

RUN 1

Many diabetes medications, such as troglitazone and pioglitazone, are broken down by specific liver enzymes called CYP3A4, which is one of the body's main drug-processing systems. Some drugs use this enzyme alone, while others use it alongside other enzymes like CYP2C8 or CYP2C9.

Repaglinide works by telling the pancreas to release insulin faster when blood sugar is high, but because it is processed by multiple enzymes, it can interact dangerously with other drugs. For example, taking repaglinide with gemfibrozil, a cholesterol-lowering medicine, can cause dangerously high blood sugar drops because the gemfibrozil prevents the body from breaking down the repaglinide.

Nateglinide is similar to repaglinide in how it boosts insulin release, but it acts faster and lasts less time. While it has fewer major interactions than repaglinide, it can still be affected by drugs that speed up or slow down the activity of the CYP3A4 and CYP2C9 enzymes.

1027 ms to first token · 6662 ms · 953 chars · 3 lines

RUN 2

Many diabetes medications, such as troglitazone and pioglitazone, are broken down by specific liver enzymes called CYP3A4, which is one of the body's main drug-processing systems. Some drugs use this enzyme alone, while others use it alongside other enzymes like CYP2C8 or CYP2C9.

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Chrome Prompt API

RUN 1

The passage discusses how certain oral medications for diabetes, called antidiabetics, are processed by the liver. These medications, like troglitazone, linagliptin, and pioglitazone, are metabolized by enzymes, specifically CYP3A4 and CYP2C8, which are like chemical processing units in the liver. Sometimes one enzyme is the main processor, while others help in smaller amounts. This means that taking other drugs that affect these enzymes can change how much of the diabetes medication is in your bloodstream, potentially leading to either too much or too little medication.

Repaglinide and nateglinide are examples of these diabetes medications. Repaglinide works by stimulating insulin release from the pancreas by affecting potassium channels in pancreatic cells. Nateglinide does something similar, but acts more quickly and has a shorter duration of action. Both drugs are processed by CYP2C8 and CYP3A4, making them susceptible to interactions with other medications. For instance, taking certain drugs that inhibit these enzymes can increase the concentration of repaglinide in the blood, increasing the risk of hypoglycemia (low blood sugar). Conversely, drugs that speed up the breakdown of these enzymes can reduce the amount of repaglinide in the blood, potentially making it less effective.

The passage highlights the importance of understanding how these medications are metabolized to avoid adverse drug interactions. For example, combining repaglinide with gemfibrozil (a cholesterol-lowering drug) can lead to dangerously low blood sugar. Similarly, rifampicin (an antibiotic) can reduce the effectiveness of nateglinide. The interplay dangerously high blood sugar drops because the gemfibrozil prevents the body from breaking down the repaglinide.

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`1012 ms to first token · 6603 ms · 953 chars · 3 lines`

`identical across runs`

between these metabolic pathways and the presence of transporters, which help move drugs into and out of cells, further complicates the picture. Therefore, healthcare providers need to consider all these factors when prescribing medications to patients with diabetes.

`912 ms to first token · 6948 ms · 1932 chars · 3 lines`

RUN 2

The passage discusses how certain oral medications for diabetes, called antidiabetics, are processed by the liver. These medications, like troglitazone, linagliptin, and pioglitazone, are metabolized by enzymes, specifically CYP3A4 and CYP2C8, which are like chemical processing units in the liver. Sometimes one enzyme is the main processor, while others help in smaller amounts. This means that taking other drugs that affect these enzymes can change how much of the diabetes medication is in your bloodstream, potentially leading to either too much or too little medication.

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745 ms to first token · 6747 ms · 1932 chars · 3 lines

identical across runs

5.1. Lipid-Lowering Drugs and CYP3A4 RECAP → PAGE LANGUAGE

(unclassified capture)

LiteRT-LM · Gemma 4 E4B

RUN 1

Most statins are substrates of CYP3A4, making them vulnerable to pharmacokinetic interactions.
Inhibition of CYP3A4 can elevate plasma statin concentrations, increasing the risk of myopathy.
Simvastatin bioavailability is less than 5% due to extensive CYP3A4 metabolism.
Cerivastatin achieves peak plasma concentrations within 2 to 3 hours after oral intake.

1121 ms to first token · 3134 ms · 358 chars · 4 lines

RUN 2

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Inhibition of CYP3A4 can elevate plasma statin concentrations, increasing the risk of myopathy.
Simvastatin bioavailability is less than 5% due to extensive CYP3A4 metabolism.
Cerivastatin achieves peak plasma concentrations within 2 to 3 hours after oral intake.

907 ms to first token · 2868 ms · 358 chars · 4 lines

identical across runs

Chrome Prompt API

RUN 1

Statins are often metabolized by CYP3A4, leading to potential drug interactions if CYP3A4 is inhibited or induced.
Simvastatin bioavailability is less than 5% due to extensive first-pass metabolism via CYP3A4, making it susceptible to interactions with CYP3A4 inhibitors like ketoconazole.
Cerivastatin is rapidly absorbed and achieves dose-proportional increases in plasma concentration, primarily accumulating in hepatic tissue.
Antidiabetic drugs are also metabolized by CYP3A4, and interactions can significantly alter their efficacy, especially in patients with multiple comorbidities.

897 ms to first token · 2784 ms · 594 chars · 4 lines

RUN 2

Statins are often metabolized by CYP3A4, leading to potential drug interactions if CYP3A4 is inhibited or induced.
Simvastatin bioavailability is less than 5% due to extensive first-pass metabolism via CYP3A4, making it susceptible to interactions with CYP3A4 inhibitors like ketoconazole.
Cerivastatin is rapidly absorbed and achieves dose-proportional increases in plasma concentration, primarily accumulating in hepatic tissue.
Antidiabetic drugs are also metabolized by CYP3A4, and interactions can significantly alter their efficacy, especially in patients with multiple comorbidities.

807 ms to first token · 2753 ms · 594 chars · 4 lines

identical across runs

5.1. Lipid-Lowering Drugs and CYP3A4 FEYNMAN → PAGE LANGUAGE

(unclassified capture)

LiteRT-LM · Gemma 4 E4B

RUN 1

Many cholesterol-lowering drugs, known as statins, rely on an enzyme called CYP3A4 to break them down after they are taken orally. If another medication blocks this enzyme, the statin builds up in the bloodstream, which can cause serious muscle problems. Conversely, if another drug speeds up the enzyme, the statin might not work as well.

Specific statins like simvastatin and lovastatin are heavily processed by CYP3A4 in the liver and intestines, meaning very little of the original drug actually reaches the bloodstream. This makes them highly sensitive to inhibitors like ketoconazole, which can dangerously increase their levels.

For diabetes medications, the same CYP3A4 enzyme plays a role in how they are processed, making them susceptible to similar interactions. This is a major concern because people with diabetes often take several different medications for other health issues, increasing the chance of these drug interactions occurring.

1111 ms to first token · 5732 ms · 954 chars · 3 lines

RUN 2

Many cholesterol-lowering drugs, known as statins, rely on an enzyme called CYP3A4 to break them down after they are taken orally. If another medication blocks this enzyme, the statin builds up in the bloodstream, which can cause serious muscle problems. Conversely, if another drug speeds up the enzyme, the statin might not work as well.

Specific statins like simvastatin and lovastatin are heavily processed by CYP3A4 in the liver and

Chrome Prompt API

RUN 1

Statins, which are commonly used to lower cholesterol, are metabolized by an enzyme called CYP3A4. CYP3A4 is a liver enzyme that helps break down many substances in the body. When other drugs interfere with CYP3A4, it can affect how much of a statin remains in the bloodstream. If CYP3A4 is inhibited, statin levels can increase, potentially leading to side effects like muscle problems. Conversely, if CYP3A4 is stimulated, statin levels can decrease, reducing the drug's effectiveness.

Specific statins like simvastatin, atorvastatin, lovastatin, and cerivastatin all rely on CYP3A4 for their breakdown. Simvastatin, for example, has very low bioavailability, meaning only a small amount of the drug actually reaches the bloodstream, making it highly sensitive to CYP3A4 inhibitors. Atorvastatin and lovastatin also have significant amounts of their metabolism occurring through CYP3A4, and therefore are susceptible to interactions with drugs that affect this enzyme. Cerivastatin, however, is absorbed very well and has a predictable pharmacokinetic profile.

Antidiabetic drugs, used to manage conditions like type 2 diabetes, also rely on CYP3A4 for their metabolism. Changes in CYP3A4 activity can significantly impact how these drugs work in the body. This is particularly important because many diabetic patients take multiple medications to manage their overall health, increasing the likelihood of drug interactions and variations in how individuals respond to treatment.

806 ms to first token · 5952 ms · 1483 chars · 3 lines

RUN 2

intestines, meaning very little of the original drug actually reaches the bloodstream. This makes them highly sensitive to inhibitors like ketoconazole, which can dangerously increase their levels.

For diabetes medications, the same CYP3A4 enzyme plays a role in how they are processed, making them susceptible to similar interactions. This is a major concern because people with diabetes often take several different medications for other health issues, increasing the chance of these drug interactions occurring.

920 ms to first token · 5504 ms · 954 chars · 3 lines

identical across runs

Statins, which are commonly used to lower cholesterol, are metabolized by an enzyme called CYP3A4. CYP3A4 is a liver enzyme that helps break down many substances in the body. When other drugs interfere with CYP3A4, it can affect how much of a statin remains in the bloodstream. If CYP3A4 is inhibited, statin levels can increase, potentially leading to side effects like muscle problems. Conversely, if CYP3A4 is stimulated, statin levels can decrease, reducing the drug's effectiveness.

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1651 ms to first token · 7600 ms · 1483 chars · 3 lines

identical across runs

MILESTONES: 1969–1976 RECAP → PAGE LANGUAGE (unclassified capture)

LiteRT-LM · Gemma 4 E4B

RUN 1

President Richard M. Nixon announced his New Economic Policy on August 15, 1971.
The Bretton Woods system fixed foreign currency values to the U.S. dollar, which was pegged to gold at $35 per ounce.
Nixon's policy included suspending the dollar's convertibility into gold and imposing a 10 percent tariff on imports.
The New Economic Policy aimed to create jobs, stop the cost of living from rising, and protect the dollar from speculators.

2157 ms to first token · 4885 ms · 440 chars · 4 lines

RUN 2

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The Bretton Woods system fixed foreign currency values to the U.S. dollar, which was pegged to gold at $35 per ounce.
Nixon's policy included suspending the dollar's convertibility into gold and imposing a 10 percent tariff on imports.
The New Economic Policy aimed to create jobs, stop the cost of living from rising, and protect the dollar from speculators.

1991 ms to first token · 5134 ms · 440 chars · 4 lines

identical across runs

Chrome Prompt API

RUN 1

President Nixon announced the New Economic Policy on August 15, 1971, marking the end of the Bretton Woods system.
The Bretton Woods system fixed foreign currency values to the U.S. dollar, which was linked to gold at $35 per ounce.
A surplus of U.S. dollars threatened the Bretton Woods system due to insufficient gold reserves.
Nixon suspended the dollar's convertibility into gold and levied a 10% tariff on imports to encourage currency adjustments.

1484 ms to first token · 3925 ms · 457 chars · 4 lines

RUN 2

President Nixon announced the New Economic Policy on August 15, 1971, marking the end of the Bretton Woods system.
The Bretton Woods system fixed foreign currency values to the U.S. dollar, which was linked to gold at $35 per ounce.
A surplus of U.S. dollars threatened the Bretton Woods system due to insufficient gold reserves.
Nixon suspended the dollar's convertibility into gold and levied a 10% tariff on imports to encourage currency adjustments.

1703 ms to first token · 3953 ms · 457 chars · 4 lines

identical across runs

MILESTONES: 1969–1976 FEYNMAN → PAGE LANGUAGE

(unclassified capture)

LiteRT-LM · Gemma 4 E4B

RUN 1

The Bretton Woods system was an international agreement established after World War II that kept the value of different countries' currencies tied to the U.S. dollar, which was itself backed by a fixed amount of gold. By the 1960s, the U.S. had too many dollars circulating globally, which made the dollar appear more valuable than it actually was, causing problems for the American economy.

Because the dollar was overvalued, international traders began selling it, creating instability. To fix this, President Richard M. Nixon announced a new economic policy in 1971, which included suspending the ability to exchange U.S. dollars for gold. He also proposed tax cuts and price freezes to boost the economy and protect the dollar from speculators.

2488 ms to first token · 7234 ms · 749 chars · 2 lines

RUN 2

The Bretton Woods system was an international agreement established after World War II that kept the value of different countries' currencies tied to the U.S. dollar, which was itself backed by a fixed amount of gold. By the 1960s, the U.S. had too many dollars circulating globally, which made the dollar appear more valuable than it actually was, causing problems for the American economy.

Because the dollar was overvalued, international traders began selling it, creating instability. To fix this, President Richard M. Nixon announced a new economic policy in 1971, which included suspending the ability to exchange U.S. dollars for gold. He also proposed tax cuts and price freezes to boost the economy and protect the dollar from speculators.

Chrome Prompt API

RUN 1

The passage describes a pivotal moment in international economics in 1971. The Bretton Woods system was a set of agreements established after World War II that linked the value of different countries' currencies to the U.S. dollar, which in turn was linked to gold. This system was meant to create stability in global trade. However, by the 1960s, the U.S. was running a surplus of dollars, meaning there were more dollars in circulation than gold to back them, making the dollar overvalued.

President Nixon responded to this situation with what became known as the "Nixon shock." He announced a new economic policy that essentially ended the Bretton Woods system. He suspended the dollar's convertibility to gold, meaning other countries could no longer exchange dollars for gold. He also implemented a 10% tariff on imports, hoping to encourage other countries to revalue their currencies and lower their trade barriers. The goal was to stabilize the dollar and protect the U.S. from what he called "attacks of international money speculators."

Nixon's actions were prompted by concerns that the overvalued dollar was harming American trade. He convened a meeting with his economic advisors to discuss a course of action, ultimately deciding to decouple the dollar from gold and implement other measures to control inflation and stimulate the economy. This decision marked a significant shift in the global financial landscape, moving away from a system of fixed exchange rates towards a more flexible one.

2404 ms to first token · 6774 ms · 749 chars · 2 lines

identical across runs

1873 ms to first token · 9156 ms · 1516 chars · 3 lines

RUN 2

The passage describes a pivotal moment in international economics in 1971. The Bretton Woods system was a set of agreements established after World War II that linked the value of different countries' currencies to the U.S. dollar, which in turn was linked to gold. This system was meant to create stability in global trade. However, by the 1960s, the U.S. was running a surplus of dollars, meaning there were more dollars in circulation than gold to back them, making the dollar overvalued.

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1658 ms to first token · 8949 ms · 1516 chars · 3 lines

	identical across runs

6.2.Native Performance RECAP → PAGE LANGUAGE (unclassified capture)

LiteRT-LM · Gemma 4 E4B

RUN 1

The three GPU clusters represent high-end discrete, integrated/mobile, and low-power devices.
High-end discrete GPUs like the NVIDIA RTX 5080 are in the first performance cluster.
Low-power GPUs in the third cluster could only run the four smallest models.
On high-cluster devices, smaller models achieve over 3k tok/s during prefill.

1876 ms to first token · 4464 ms · 334 chars · 4 lines

RUN 2

The three GPU clusters represent high-end discrete, integrated/mobile, and low-power devices.
High-end discrete GPUs like the NVIDIA RTX 5080 are in the first performance cluster.
Low-power GPUs in the third cluster could only run the four smallest models.
On high-cluster devices, smaller models achieve over 3k tok/s during prefill.

1757 ms to first token · 4280 ms · 334 chars · 4 lines

identical across runs

Chrome Prompt API

RUN 1

LlamaWeb was evaluated on 10 models across 16 GPUs from 8 vendors, grouped into three performance clusters.
The clusters represent GPU capabilities: high-end discrete GPUs, integrated/high-end mobile GPUs, and low-power/efficient GPUs from iPhones and Android devices.
Smaller models achieve over 3k tok/s prefill and 100 tok/s decode on high-cluster devices, while larger models drop to 65 tok/s prefill and 30 tok/s decode.
LlamaWeb scales to handle higher contexts while maintaining reasonable performance, even on constrained devices.

1049 ms to first token · 4024 ms · 542 chars · 4 lines

RUN 2

LlamaWeb was evaluated on 10 models across 16 GPUs from 8 vendors, grouped into three performance clusters.
The clusters represent GPU capabilities: high-end discrete GPUs, integrated/high-end mobile GPUs, and low-power/efficient GPUs from iPhones and Android devices.
Smaller models achieve over 3k tok/s prefill and 100 tok/s decode on high-cluster devices, while larger models drop to 65 tok/s prefill and 30 tok/s decode.
LlamaWeb scales to handle higher contexts while maintaining reasonable performance, even on constrained devices.

1609 ms to first token · 4168 ms · 542 chars · 4 lines

	identical across runs

6.2.Native Performance FEYNMAN → PAGE LANGUAGE (unclassified capture)

LiteRT-LM · Gemma 4 E4B

RUN 1

The researchers tested how well a system called LlamaWeb, which runs large language models in a web browser, performs across many different types of computer graphics cards (GPUs). They grouped these GPUs into three performance levels: high-end desktop cards, mid-range mobile and integrated cards, and low-power cards found in phones.

They measured two types of speed: "prefill throughput," which is how fast the system processes the initial prompt you type, and "decode throughput," which is how fast it generates the response word by word. The results show that while high-end cards are much faster, the system still works on low-power phones, though much slower.

This testing demonstrates that LlamaWeb is flexible enough to run various AI models on a wide range of hardware, from powerful desktop computers to small mobile devices, even when memory is limited.

2008 ms to first token · 6517 ms · 867 chars · 3 lines

RUN 2

The researchers tested how well a system called LlamaWeb, which runs large language models in a web browser, performs across many different types of computer graphics cards (GPUs). They grouped these GPUs into three performance levels: high-end desktop cards, mid-range mobile and integrated cards, and low-power cards found in phones.

They measured two types of speed: "prefill throughput," which is how fast the system processes the initial prompt you type, and "decode throughput," which is how fast it

Chrome Prompt API

RUN 1

The study evaluated how well the LlamaWeb inference engine works across different devices. They tested ten different language models on a wide range of GPUs from various manufacturers, grouping them into three performance clusters based on how they perform. This grouping helps understand the capabilities of different GPUs, with the high cluster containing powerful GPUs like the NVIDIA RTX 5080, the mid cluster including integrated and mobile GPUs like the Apple M2, and the low cluster featuring less powerful GPUs found in iPhones and Android devices.

The researchers measured the "throughput" of LlamaWeb, which is the speed at which the engine processes text. They measured two types of throughput: "prefill throughput" (how quickly the engine prepares for generating text) and "decode throughput" (how quickly the engine generates text). The results showed that throughput varied greatly depending on the GPU's performance and the size of the language model. Smaller models performed better on lower-end GPUs, while larger models required more powerful hardware.

The study also examined how different software "backends" and "weight formats" affected performance. A backend is the software that allows the model to run on a specific type of GPU, and weight formats refer to how the model's parameters are stored. The results indicated that while these factors can influence performance, LlamaWeb is adaptable and can still achieve reasonable speeds even on less powerful devices.

909 ms to first token · 6323 ms · 1489 chars · 3 lines

RUN 2

The study evaluated how well the LlamaWeb inference engine works across different devices.

generates the response word by word. The results show that while high-end cards are much faster, the system still works on low-power phones, though much slower.

This testing demonstrates that LlamaWeb is flexible enough to run various AI models on a wide range of hardware, from powerful desktop computers to small mobile devices, even when memory is limited.

1969 ms to first token · 6365 ms · 867 chars · 3 lines

identical across runs

They tested ten different language models on a wide range of GPUs from various manufacturers, grouping them into three performance clusters based on how they perform. This grouping helps understand the capabilities of different GPUs, with the high cluster containing powerful GPUs like the NVIDIA RTX 5080, the mid cluster including integrated and mobile GPUs like the Apple M2, and the low cluster featuring less powerful GPUs found in iPhones and Android devices.

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1085 ms to first token · 6054 ms · 1489 chars · 3 lines

identical across runs

3.Design of LlamaWeb RECAP → PAGE LANGUAGE (unclassified capture)

LiteRT-LM · Gemma 4 E4B

RUN 1

Kernels are compiled at runtime into pipelines that define executable code and resource layouts.
WebGPU uses a deferred execution model where compute workloads are recorded and flushed into command buffers.
LlamaWeb allocates a single buffer at startup to manage dynamic kernel parameters without synchronization logic.
LlamaWeb used significantly less memory than Transformers.js and WebLLM across both Linux and Windows tests.

1758 ms to first token · 3712 ms · 428 chars · 4 lines

RUN 2

Kernels are compiled at runtime into pipelines that define executable code and resource layouts.
WebGPU uses a deferred execution model where compute workloads are recorded and flushed into command buffers.
LlamaWeb allocates a single buffer at startup to manage dynamic kernel parameters without synchronization logic.
LlamaWeb used significantly less memory than Transformers.js and WebLLM across both Linux and Windows tests.

1106 ms to first token · 3150 ms · 428 chars · 4 lines

identical across runs

Chrome Prompt API

RUN 1

Applications use a structured API to interact with the GPU, compiling kernels into pipelines.
WebGPU uses a deferred execution model, recording workloads via command encoders and passes.
WebGPU kernels require dynamic parameters, and WebLLM uses cached buffers to pass data to kernels at runtime.
LlamaWeb uses less memory than Transformers.js and WebLLM on NVIDIA RTX 5080, demonstrating its memory-efficient design.

1073 ms to first token · 2713 ms · 421 chars · 4 lines

RUN 2

Applications use a structured API to interact with the GPU, compiling kernels into pipelines.
WebGPU uses a deferred execution model, recording workloads via command encoders and passes.
WebGPU kernels require dynamic parameters, and WebLLM uses cached buffers to pass data to kernels at runtime.
LlamaWeb uses less memory than Transformers.js and WebLLM on NVIDIA RTX 5080, demonstrating its memory-efficient design.

1399 ms to first token · 2913 ms · 421 chars · 4 lines

identical across runs

3.Design of LlamaWeb FEYNMAN → PAGE LANGUAGE (unclassified capture)

LiteRT-LM · Gemma 4 E4B

RUN 1

The passage describes how software applications talk to a graphics processing unit (GPU) using a specific set of rules called an API. Instead of running code immediately, the system records instructions into a command buffer, which is then sent to the GPU queue to be executed later. This recording and submission process is how the application tells the GPU what to do, including tasks like moving data between memory locations.

Because the system doesn't easily allow passing small, changing pieces of information to the GPU code at runtime, the developers created a clever workaround. They pre-allocate one large block of memory at the start and reuse slots within it for different tasks, ensuring that one task's data doesn't get overwritten before it's finished. This design keeps things organized and prevents the system from running out of memory.

The text also details how specific mathematical operations, like matrix multiplication and attention mechanisms used in large language models, are optimized to use less memory. By processing data in small chunks and keeping necessary calculations local to the GPU's processing units, the system achieves significant memory savings compared to other popular frameworks.

1393 ms to first token · 6906 ms · 1225 chars · 3 lines

RUN 2

The passage describes how software applications talk to a graphics processing unit (GPU) using a specific set of rules called an API. Instead of running code immediately, the system records instructions into a command buffer, which is then sent to the GPU queue to be executed later. This recording and submission process is how the application tells the GPU what to do, including

Chrome Prompt API

RUN 1

WebGPU is a way for applications to use the graphics processing unit (GPU) for calculations. Applications communicate with the GPU through a specific programming interface, and the instructions for these calculations, called kernels, are compiled into pipelines. These pipelines define both the code to be executed and how data is organized on the GPU. Data is stored in buffers, which are like temporary storage areas, and these buffers are grouped into bind groups, which are then linked to the pipelines. WebGPU uses a "deferred execution" model, meaning that the GPU doesn't start working on tasks until a command encoder, which packages the instructions, is finished and submitted to the GPU.

A challenge with WebGPU is managing small amounts of data needed for each kernel, like matrix dimensions. Currently, WebGPU doesn't offer a direct way to pass this data at runtime. To overcome this, frameworks like WebLLM and ONNX Runtime use "buffer pools," which are collections of pre-allocated buffers. These pools are dynamically allocated and cached to avoid repeatedly allocating memory, which can slow things down or even cause crashes if the GPU runs out of memory. LlamaWeb addresses this by allocating a single buffer at startup with enough space for the necessary parameters, rotating through the slots in the buffer as needed, and ensuring parameters aren't overwritten until the kernel using them is finished. This approach avoids the need for complex synchronization mechanisms and reduces memory fragmentation.

LlamaWeb focuses on optimizing kernels for tasks like matrix multiplication and attention, a key operation in large language models (LLMs). The attention kernels are designed to efficiently tasks like moving data between memory locations.

Because the system doesn't easily allow passing small, changing pieces of information to the GPU code at runtime, the developers created a clever workaround. They pre-allocate one large block of memory at the start and reuse slots within it for different tasks, ensuring that one task's data doesn't get overwritten before it's finished. This design keeps things organized and prevents the system from running out of memory.

The text also details how specific mathematical operations, like matrix multiplication and attention mechanisms used in large language models, are optimized to use less memory. By processing data in small chunks and keeping necessary calculations local to the GPU's processing units, the system achieves significant memory savings compared to other popular frameworks.

1284 ms to first token · 6842 ms · 1225 chars · 3 lines

identical across runs

process the KV cache, which stores information used in attention calculations. Different versions of the attention kernel are implemented, including FlashAttention, which streams over the KV cache in tiles to improve performance. LlamaWeb also supports quantized KV-cache formats, which reduce memory usage by using lower-precision data. Performance tests show that LlamaWeb uses significantly less memory than other frameworks like Transformers.js and WebLLM, especially on Windows, and avoids memory leaks, highlighting the importance of memory management when developing WebGPU applications.

1096 ms to first token · 8606 ms · 2315 chars · 3 lines

RUN 2

WebGPU is a way for applications to use the graphics processing unit (GPU) for calculations. Applications communicate with the GPU through a specific programming interface, and the instructions for these calculations, called kernels, are compiled into pipelines. These pipelines define both the code to be executed and how data is organized on the GPU. Data is stored in buffers, which are like temporary storage areas, and these buffers are grouped into bind groups, which are then linked to the pipelines. WebGPU uses a "deferred execution" model, meaning that the GPU doesn't start working on tasks until a command encoder, which packages the instructions, is finished and submitted to the GPU.

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LlamaWeb focuses on optimizing kernels for tasks like matrix multiplication and attention, a key operation in large language models (LLMs). The attention kernels are designed to efficiently process the KV cache, which stores information used in attention calculations. Different versions of the attention kernel are implemented, including FlashAttention, which streams over the KV cache in tiles to improve performance. LlamaWeb also supports quantized KV-cache formats, which reduce memory usage by using lower-precision data. Performance tests show that LlamaWeb uses significantly less memory than other frameworks like Transformers.js and WebLLM, especially on Windows, and avoids memory leaks, highlighting the importance of memory management when developing WebGPU applications.

1300 ms to first token · 8510 ms · 2315 chars · 3 lines

identical across runs

Autism and Reading Part 2: Lessons to be Learned from Special Kids RECAP → PAGE LANGUAGE

(unclassified capture)

LiteRT-LM · Gemma 4 E4B

RUN 1

Studies show that ASD students have difficulty understanding the mental states of others.
Typical reading tests often fail to address social insensitivity or emotional comprehension.
A suggested instructional method involves modeling emotional reactions in stories before guiding students to infer them.
Research has shown some positive steps in teaching students how to use cohesive links in text.

1473 ms to first token · 3128 ms · 398 chars · 4 lines

RUN 2

Studies show that ASD students have difficulty understanding the mental states of others.
Typical reading tests often fail to address social insensitivity or emotional comprehension.
A suggested instructional method involves modeling emotional reactions in stories before guiding students to infer them.
Research has shown some positive steps in teaching students how to use cohesive links in text.

1095 ms to first token · 2725 ms · 398 chars · 4 lines

identical across runs

Chrome Prompt API

RUN 1

Studies treat ASD like other reading disabilities by increasing the intensity of existing successful methods.
Individuals with ASD often struggle with understanding the mental states of others, a concept called "theory of mind."
Typical reading tests often focus on literal recall and inferencing, missing the core reasons for students' errors.
Teaching emotional reactions in stories and guiding students to infer character emotions could be beneficial.
Some small research suggests that teaching cohesive links and building understanding of a text step-by-step may help.

949 ms to first token · 2640 ms · 576 chars · 5 lines

RUN 2

Studies treat ASD like other reading disabilities by increasing the intensity of existing successful methods.
Individuals with ASD often struggle with understanding the mental states of others, a concept called "theory of mind."
Typical reading tests often focus on literal recall and inferencing, missing the core reasons for students' errors.
Teaching emotional reactions in stories and guiding students to infer character emotions could be beneficial.
Some small research suggests that teaching cohesive links and building understanding of a text step-by-step may help.

	1200 ms to first token · 2961 ms · 576 chars · 5 lines identical across runs

Autism and Reading Part 2: Lessons to be Learned from Special Kids FEYNMAN → PAGE LANGUAGE

(unclassified capture)

LiteRT-LM · Gemma 4 E4B

RUN 1

Some studies try to help people with Autism Spectrum Disorder (ASD) by simply giving them more of the same reading lessons that work for others, like using more one-on-one teaching or more practice. While these methods sometimes show positive results, it is unclear if they would work as well in a typical classroom setting.

A major challenge for people with ASD is "theory of mind," which is the ability to understand that other people have different thoughts and feelings than they do. Because of this difficulty, they often struggle to understand social situations and the intentions behind what others do, which makes empathy hard.

Standard reading tests often ask questions about a character's feelings or reasons for acting, but these questions are usually labeled as simple comprehension tasks. The author argues that these tests miss the real issue, which is the difficulty in understanding social emotions, and suggests a better approach: teaching students how to recognize and infer emotional reactions in stories.

1421 ms to first token · 6046 ms · 1026 chars · 3 lines

RUN 2

Some studies try to help people with Autism Spectrum Disorder (ASD) by simply giving them more of the same reading lessons that work for others, like using more one-on-one teaching or more practice. While these methods sometimes show positive results, it is unclear if they would work as well in a typical classroom setting.

A major challenge for people with ASD is "theory of mind," which is the ability to understand that

Chrome Prompt API

RUN 1

Research suggests that students with Autism Spectrum Disorder (ASD) often struggle with understanding the inner lives of others, a concept known as "theory of mind." This means they may find it difficult to grasp the feelings, intentions, and motivations behind people's actions, which can contribute to social challenges and difficulties with empathy. Traditional reading assessments often focus on factual recall and comprehension skills, but they don't adequately address these social understanding deficits.

Instead of focusing solely on traditional reading skills, a more effective approach might involve explicitly teaching students about emotions and how they manifest in stories. This could involve modeling emotional responses to characters, guiding students to infer feelings, and exploring the causes of emotions. Using graphic organizers to map character motivations and emotional reactions could also be beneficial.

While research is still developing, some promising avenues include teaching students to identify and use "cohesive links" – the connections between sentences and ideas in a text – and to build a step-by-step understanding of a passage by connecting each sentence to the previous ones. These targeted interventions, though nascent, offer a potential path towards improving reading comprehension in students with ASD by addressing their unique cognitive strengths and challenges.

794 ms to first token · 4614 ms · 1416 chars · 3 lines

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Standard reading tests often ask questions about a character's feelings or reasons for acting, but these questions are usually labeled as simple comprehension tasks. The author argues that these tests miss the real issue, which is the difficulty in understanding social emotions, and suggests a better approach: teaching students how to recognize and infer emotional reactions in stories.

1126 ms to first token · 5723 ms · 1026 chars · 3 lines

identical across runs

RUN 2

Research suggests that students with Autism Spectrum Disorder (ASD) often struggle with understanding the inner lives of others, a concept known as "theory of mind." This means they may find it difficult to grasp the feelings, intentions, and motivations behind people's actions, which can contribute to social challenges and difficulties with empathy. Traditional reading assessments often focus on factual recall and comprehension skills, but they don't adequately address these social understanding deficits.

Instead of focusing solely on traditional reading skills, a more effective approach might involve explicitly teaching students about emotions and how they manifest in stories. This could involve modeling emotional responses to characters, guiding students to infer feelings, and exploring the causes of emotions. Using graphic organizers to map character motivations and emotional reactions could also be beneficial.

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1210 ms to first token · 5127 ms · 1416 chars · 3 lines

identical across runs

Exploring Claude/GPT Knowledge Cutoffs & Pre-training Timelines RECAP → PAGE LANGUAGE

(unclassified capture)

LiteRT-LM · Gemma 4 E4B

RUN 1

Anthropic Opus 4.7 onwards models share a knowledge cutoff around late December 2025.
Vertical lines in the graph indicate active post-training on recency-biased datasets.
OpenAI models show training patterns involving GPT-4, GPT-4o, GPT-4.1, GPT-5, and ChatGPT.
Anthropic models show training patterns involving 3.5 Sonnet and more recent Sonnet 4.5 models.

1370 ms to first token · 3767 ms · 358 chars · 4 lines

RUN 2

Anthropic Opus 4.7 onwards models share a knowledge cutoff around late December 2025.
Vertical lines in the graph indicate active post-training on recency-biased datasets.
OpenAI models show training patterns involving GPT-4, GPT-4o, GPT-4.1, GPT-5, and ChatGPT.
Anthropic models show training patterns involving 3.5 Sonnet and more recent Sonnet 4.5 models.

929 ms to first token · 3281 ms · 358 chars · 4 lines

identical across runs

Chrome Prompt API

RUN 1

Anthropic's Opus 4.7 models were trained on a dataset that cuts off around late December 2025.
Model knowledge cutoffs correlate with the dataset's time span used during pre-training.
Models often repeat information about being a specific model (e.g., GPT-4, Claude 3.5 Sonnet) due to seeing it frequently in their training data.
Labs frequently train on chat data from their respective platforms (ChatGPT.com, Claude.ai).

772 ms to first token · 2419 ms · 426 chars · 4 lines

RUN 2

Anthropic's Opus 4.7 models were trained on a dataset that cuts off around late December 2025.
Model knowledge cutoffs correlate with the dataset's time span used during pre-training.
Models often repeat information about being a specific model (e.g., GPT-4, Claude 3.5 Sonnet) due to seeing it frequently in their training data.
Labs frequently train on chat data from their respective platforms (ChatGPT.com, Claude.ai).

894 ms to first token · 2657 ms · 426 chars · 4 lines

identical across runs

Exploring Claude/GPT Knowledge Cutoffs & Pre-training Timelines FEYNMAN → PAGE LANGUAGE

(unclassified capture)

LiteRT-LM · Gemma 4 E4B

RUN 1

The passage explains how researchers can guess when large language models, like Claude and GPT, stopped learning new information by testing them on recent facts. By asking these models multiple-choice questions about daily events from Wikipedia and tracking when they start getting the answers wrong, the researchers can estimate the "knowledge cutoff"—the date up to which the model was trained.

The training process for these massive models involves several stages, starting with a huge, data-heavy pre-training phase where the model learns general knowledge. After this, teams perform "post-training" to improve the model's reasoning and problem-solving skills. Sometimes, smaller versions of these models are created by "distilling" knowledge from larger, more advanced versions.

The analysis also shows that models often reveal their training history by stating their own identity. If a model frequently sees itself described as a certain version during its training, it is more likely to claim that identity when asked. This suggests that some models are being trained on conversations users have had with previous versions of the AI.

1249 ms to first token · 6417 ms · 1142 chars · 3 lines

RUN 2

The passage explains how researchers can guess when large language models, like Claude and GPT, stopped learning new information by testing them on recent facts. By asking these models multiple-choice questions about daily events from Wikipedia and tracking when they start getting the answers wrong, the researchers can estimate the

Chrome Prompt API

RUN 1

Large language models like Claude and GPT are built through a multi-stage training process. First, a massive dataset of text is used to create a "pre-training checkpoint," which is like a foundational version of the model. This is a very resource-intensive step. Then, teams refine these base models through "post-training," which involves experiments to improve reasoning and problem-solving, often resulting in minor model updates. Smaller versions of these refined models are also created, becoming model families.

To understand when these models were trained, the author analyzed how accurately they answer questions about specific dates. By looking at the error rates, they can estimate when the models stopped absorbing new information from their training data. Anthropic's Opus models, for example, appear to be based on a training run that ended around late December 2025, suggesting their knowledge cutoff is around that time. The author notes that this cutoff is correlated with factual estimates, and vertical lines in the error rate data indicate periods of active post-training on more recent information.

Furthermore, the models' self-reported identities can offer clues about their training data. If a model frequently mentions being a specific version of a previous model, it's likely that version was heavily represented in its training data. The author observed patterns suggesting that OpenAI and Anthropic have trained on conversations from their respective platforms (ChatGPT and Claude), rather than relying solely on internal, synthetic datasets. This indicates that user interactions are playing a significant role in shaping the models' knowledge and behavior.

"knowledge cutoff"—the date up to which the model was trained.

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939 ms to first token · 6040 ms · 1142 chars · 3 lines

identical across runs

758 ms to first token · 5625 ms · 1695 chars · 3 lines

RUN 2

Large language models like Claude and GPT are built through a multi-stage training process. First, a massive dataset of text is used to create a "pre-training checkpoint," which is like a foundational version of the model. This is a very resource-intensive step. Then, teams refine these base models through "post-training," which involves experiments to improve reasoning and problem-solving, often resulting in minor model updates. Smaller versions of these refined models are also created, becoming model families.

To understand when these models were trained, the author analyzed how accurately they answer questions about specific dates. By looking at the error rates, they can estimate when the models stopped absorbing new information from their training data. Anthropic's Opus models, for example, appear to be based on a training run that ended around late December 2025, suggesting their knowledge cutoff is around that time. The author notes that this cutoff is correlated with factual estimates, and vertical lines in the error rate data indicate periods of active post-training on more recent information.

Furthermore, the models' self-reported identities can offer clues about their training data. If a model frequently mentions being a specific version of a previous model, it's likely that version was heavily represented in its training data. The author observed patterns suggesting that OpenAI and Anthropic have trained on conversations from their respective platforms (ChatGPT and Claude), rather than relying solely on internal, synthetic datasets. This indicates that user interactions are playing a significant role in shaping the models' knowledge and behavior.

961 ms to first token · 6064 ms · 1695 chars · 3 lines

identical across runs